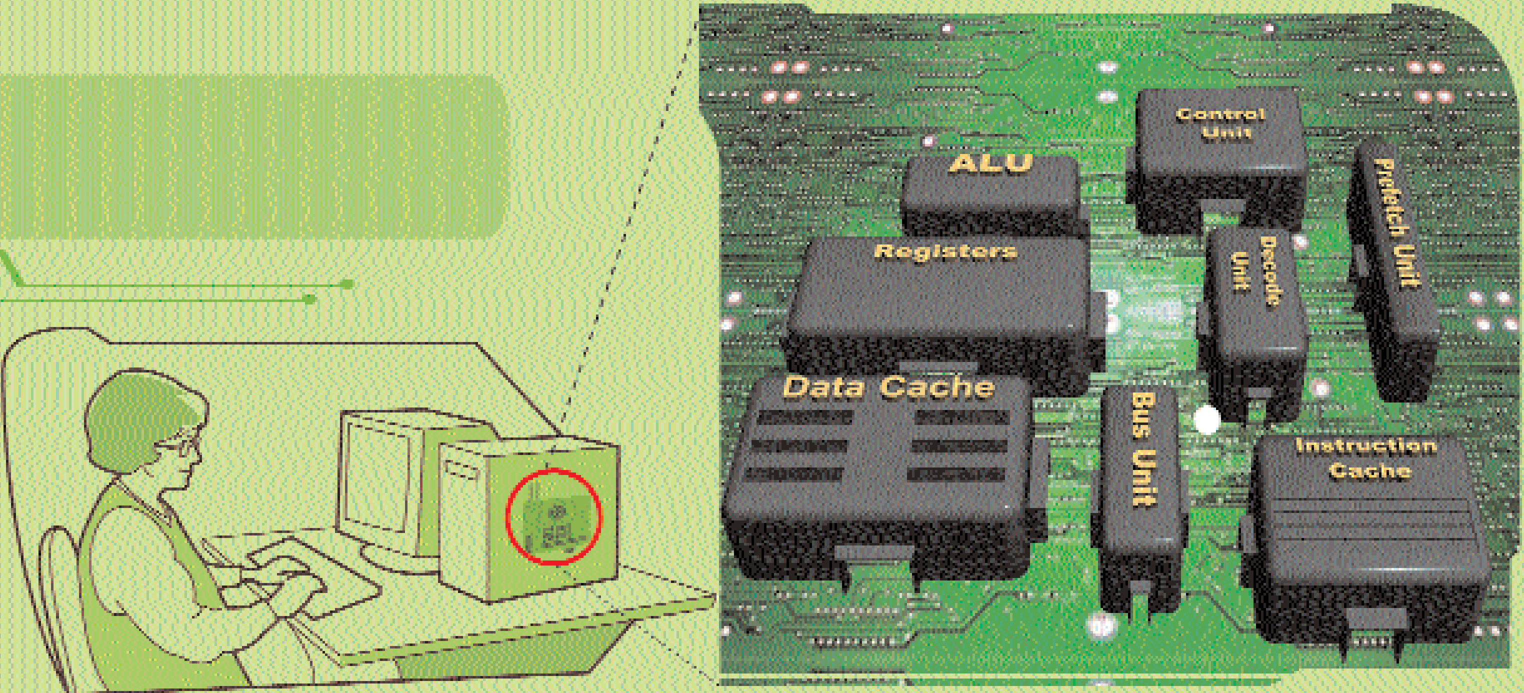

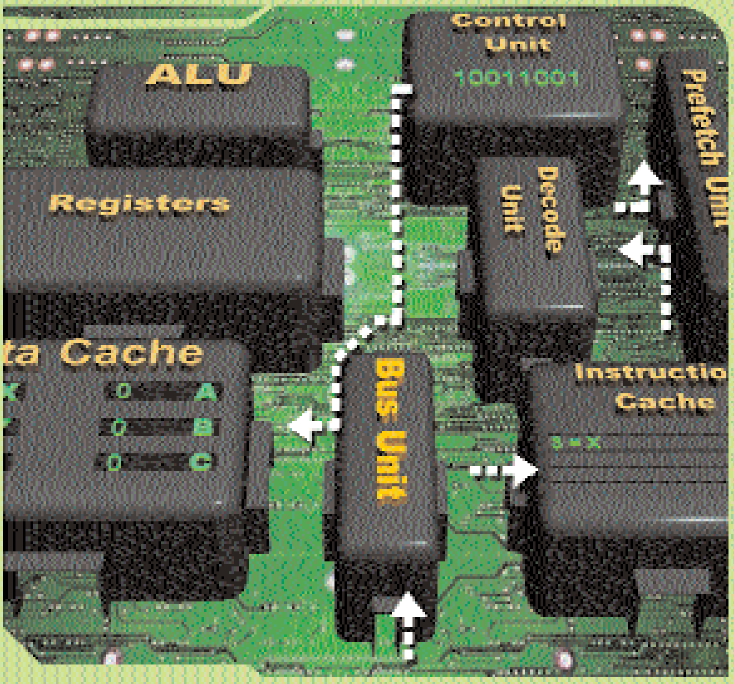

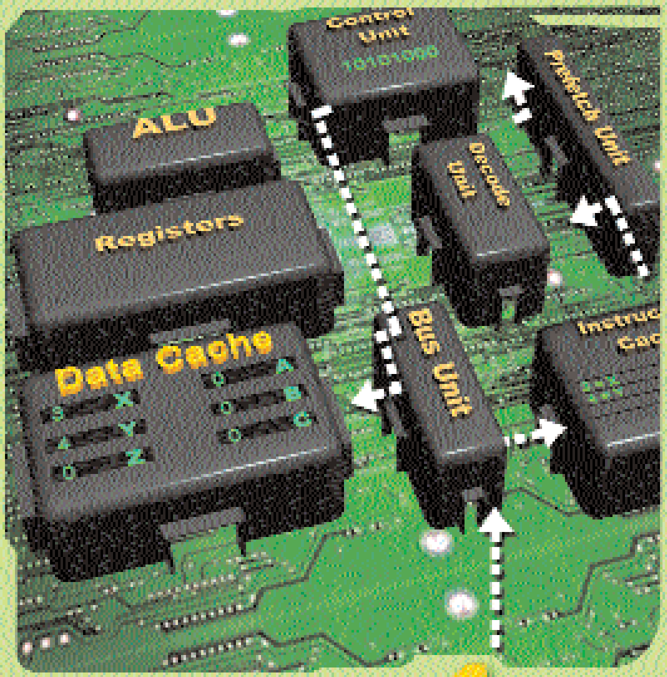

Step 1 : The number '3' is pressed

a. On pressing the '3' key, the microprocessor (µP) is alerted. It asks the Prefetch Unit to get the instruction. The new data instruction comes into the µP through the Bus Unit, and is stored in the Instruction Cache, where it is assigned 'X' as the code. So at this point, 3 equals X.

b. The Prefetch Unit sends a copy of the new data to the Decode Unit.

c. At the Decode Unit, the data is translated into binary and sent to the Control Unit and the Data Cache.

d. '3' is stored in the Data Cache for the future at an address called 'X' as per the instructions of the Control Unit.

Step 2 : The number '4' is pressed

a. On pressing the '4' key, the microprocessor (µP) is alerted. It asks the Prefetch Unit to get the instruction. The new data instruction comes into the µP through the Bus Unit, and is stored in the Instruction Cache, where it is assigned 'Y' as the code. So at this point, 4 equals Y.

b. The Prefetch Unit sends a copy of the new data to the Decode Unit.

c. At the Decode Unit, the data is translated into binary and sent to the Control Unit and the Data Cache.

d. '4' is stored in the Data Cache for the future at an address called 'Y' as per the instructions of the Control Unit.